**CME 307: Optimization** CME 307 / MS&E 311 / OIT 676

# Quasi-Newton: Practice questions

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For Exercises 1–4, let

$$f(x) = \tfrac{1}{2}x^\top Ax - b^\top x, \qquad 0 \prec \mu I \preceq A \preceq LI, \qquad \kappa = L/\mu.$$

The matrix $P$ approximates $A^{-1}$. Write $h = \nabla f(x)$ and use the full step $x^+ = x - Ph$. If $h \neq 0$, define the squared-gradient ratio $g = \|\nabla f(x^+)\|^2/\|h\|^2$. Stop if a gradient is zero.

**Exercise 1. A perfect preconditioner and a bad one.**

(a) Show that $\nabla f(x^+) = (I - AP)h$. Why does $P = A^{-1}$ solve the problem in one step?

(b) Let $A = \mathrm{diag}(1, 4)$, $b = 0$, and $x = (1, 1)^\top$. Compute $x^+$ and $g$ for $P = I$. Does this step improve the gradient norm?

(c) Why might we learn $P$ over several inexpensive updates rather than compute the perfect preconditioner first?

**Exercise 2. Repair the direction that failed.** Let $V(P) = \|I - AP\|_F^2$ and $R = I - AP$. For a unit vector $v$, suppose $r = Rv \neq 0$ and set $u = r/\|r\|$. Consider $P^+ = P + \eta uv^\top$.

(a) Expand $V(P^+)$ as a quadratic in $\eta$ and find its minimizing value $\eta^\star$.

(b) You may use the bound

$$V(P^+) \leq V(P) - \frac{1}{\kappa}\|Rv\|^2 \qquad \text{when } \eta = \eta^\star.$$

If $g > 0$, choose $v$ so that $\|Rv\|^2 = g$, and interpret the resulting decrease in $V$.

**Exercise 3. A finite budget for bad steps.** Suppose the iterates satisfy, for every $k$ before termination,

$$V_k \geq 0, \qquad V_{k+1} \leq V_k - \frac{g_k}{\kappa}, \qquad g_k = \frac{\|\nabla f(x_{k+1})\|^2}{\|\nabla f(x_k)\|^2}.$$

(a) Bound $\sum_{k=1}^{K} g_k$ in terms of $V_1$ and $\kappa$. For a fixed $\varepsilon > 0$, how many indices can have $g_k \geq \varepsilon$? What does this say about $g_k$ as $k \to \infty$ if the method never terminates?

(b) Use the arithmetic–geometric mean inequality to show

$$\frac{\|\nabla f(x_{K+1})\|^2}{\|\nabla f(x_1)\|^2} \leq \left(\frac{\kappa V_1}{K}\right)^K.$$

Explain the key idea: does an occasional step with $g_k > 1$ prevent fast convergence?

**Exercise 4. Learn from two observable gradients.** For two distinct points $x, z$, define $s = x - z$ and $y = \nabla f(x) - \nabla f(z)$.

(a) Show that $y = As$, and hence $A^{-1}y = s$. Explain why $s$ and $y$ can be formed without knowing $x^\star$.

(b) Derive the least-change update

$$P^+ = \operatorname*{argmin}_{Qy=s} \tfrac{1}{2}\|Q - P\|_F^2,$$

where $Q$ is an unrestricted real matrix. Express $P^+ - P$ as an outer product.

**Exercise 5. Learning one direction at a time.** Suppose we observe a secant pair

$$y = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad s = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}.$$

Consider two candidate inverse Hessian approximations

$$P_a = \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad P_b = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}.$$

(a) Compute $P_a y$ and $P_b y$. Do both matrices satisfy $Py = s$?

(b) Does this single secant equation determine the whole matrix $P$? Which column of $P$ does it determine?

(c) A second observation gives $\widetilde{y} = (0, 1)^\top$ and $\widetilde{s} = (0, 1/4)^\top$. Find the matrix $P$ satisfying both $Py = s$ and $P\widetilde{y} = \widetilde{s}$. What new information did the second observation provide?